Numerical Investigation of the Poisson–Boltzmann Equation

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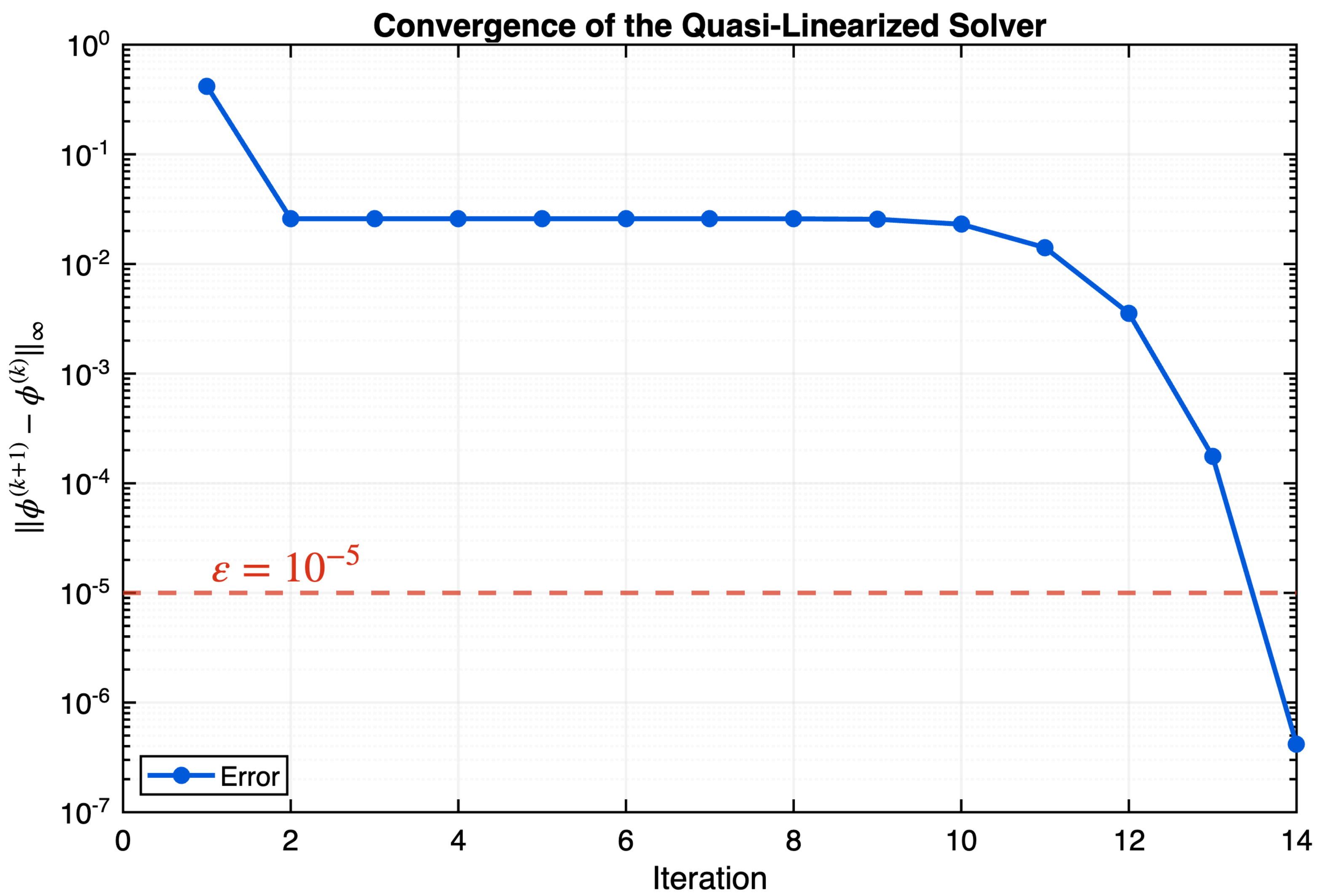
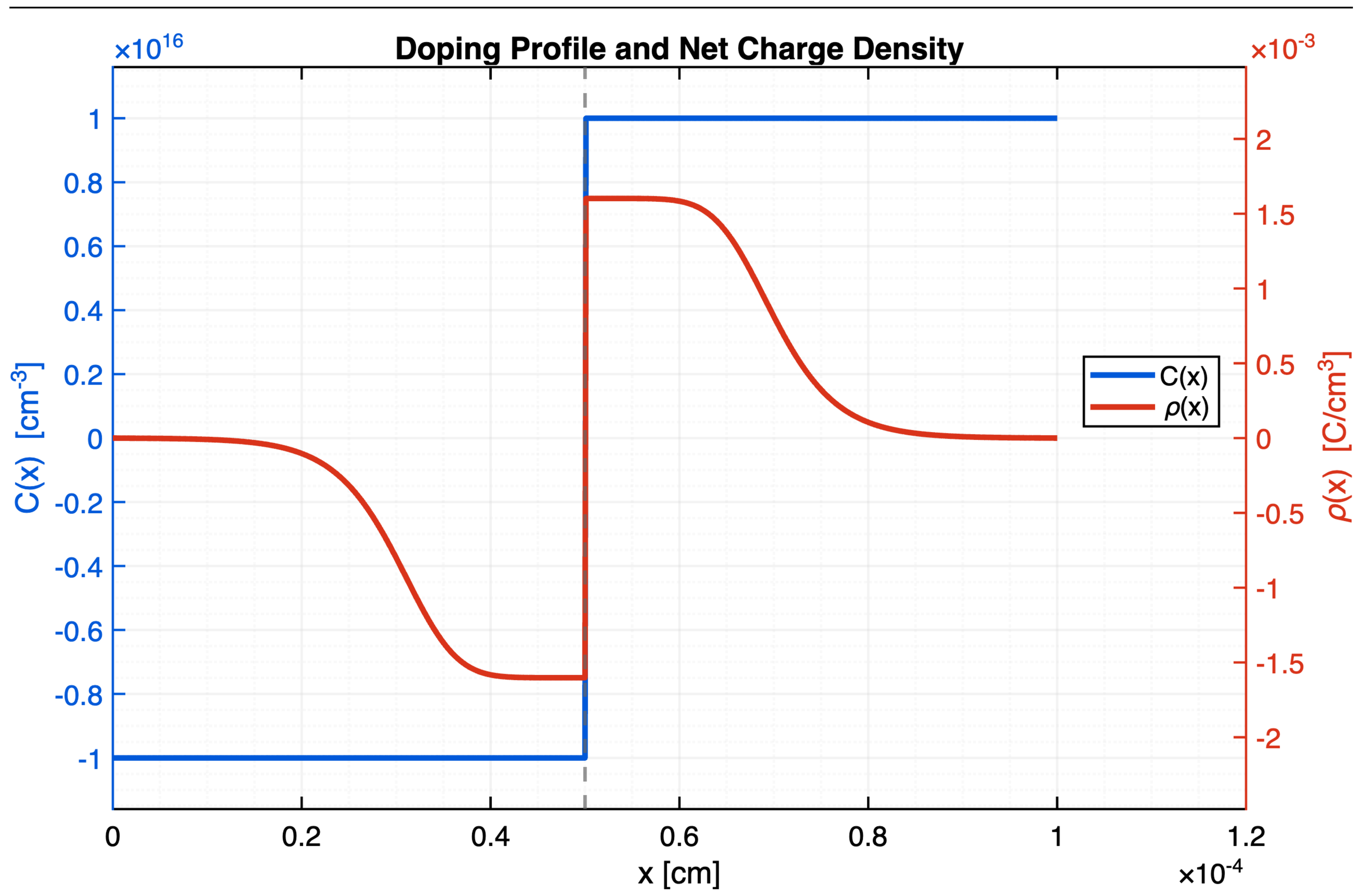
Convergence Behavior



After an initial plateau, the error decreases rapidly and falls below tolerance, $\varepsilon = 10^{-5}$ within ~ 14 iterations:

$$\|\phi^{(k+1)} - \phi^{(k)}\|_\infty < 10^{-5}.$$

Doping Profile and Net Charge Density



Net charge density, $\rho(x) = q\big(p(x) - n(x) + C(x)\big)$

The doping profile $C(x)$ determines the junction location in the device. Away from the junction, charge carriers balance the doping, so $\rho(x) \approx 0$. Near the junction, this balance breaks down, resulting in a localized region of nonzero charge. This localized imbalance directly reflects the transition between p-type and n-type regions.

Problem Statement

The electrostatic potential in semiconductor devices arises from the nonlinear interaction between charge carriers and the electric field. This leads to the **Poisson–Boltzmann equation**, whose exponential nonlinearity makes analytical solutions intractable. We implement and analyze a numerical method to compute the potential and examine its convergence behavior.

Governing Equations

Carrier densities:

$$n(x) = n_i e^{(\phi(x) - \phi_n(x))/V_T}, \qquad p(x) = n_i e^{-(\phi(x) - \phi_p(x))/V_T}$$

Poisson equation:

$$-\frac{d^2\phi}{dx^2} = \frac{q}{\varepsilon}\big(p(x) - n(x) + C(x)\big), \qquad C(x) = N_D^+(x) - N_A^-(x)$$

Poisson–Boltzmann equation at equilibrium ($\phi_n = \phi_p =$ constant):

$$-\frac{d^2\phi}{dx^2} = \frac{q}{\varepsilon}\left(n_i e^{-\phi(x)/V_T} - n_i e^{\phi(x)/V_T} + C(x)\right)$$

Hyperbolic form:

$$-\frac{d^2\phi}{dx^2} = \lambda \sinh\left(\frac{\phi(x)}{V_T}\right) - \frac{q}{\varepsilon}C(x), \qquad \lambda = \frac{2qn_i}{\varepsilon}$$

Numerical Method

Uniform grid on $[0, L]$: $x_i = ih, \quad h = \Delta x = L/m, \quad i = 0, 1, \dots, m$

$$\phi_i \approx \phi(x_i), \qquad C_i = C(x_i)$$

Discretization:

$$-\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} = \lambda \sinh\left(\frac{\phi_i}{V_T}\right) - \frac{q}{\varepsilon}C_i + \mathcal{O}(h^2)$$

Operator form:

$$A[\phi]_i := -\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2}, \qquad b[\phi]_i := \lambda \sinh\left(\frac{\phi_i}{V_T}\right) - \frac{q}{\varepsilon}C_i$$

Fixed-point iteration:

$$A[\phi^{(k+1)}] = b[\phi^{(k)}], \qquad k = 0, 1, 2, \dots; \quad \phi^{(0)} = \text{initial guess}$$

Improved method (linearization about $\phi^{(k)}$):

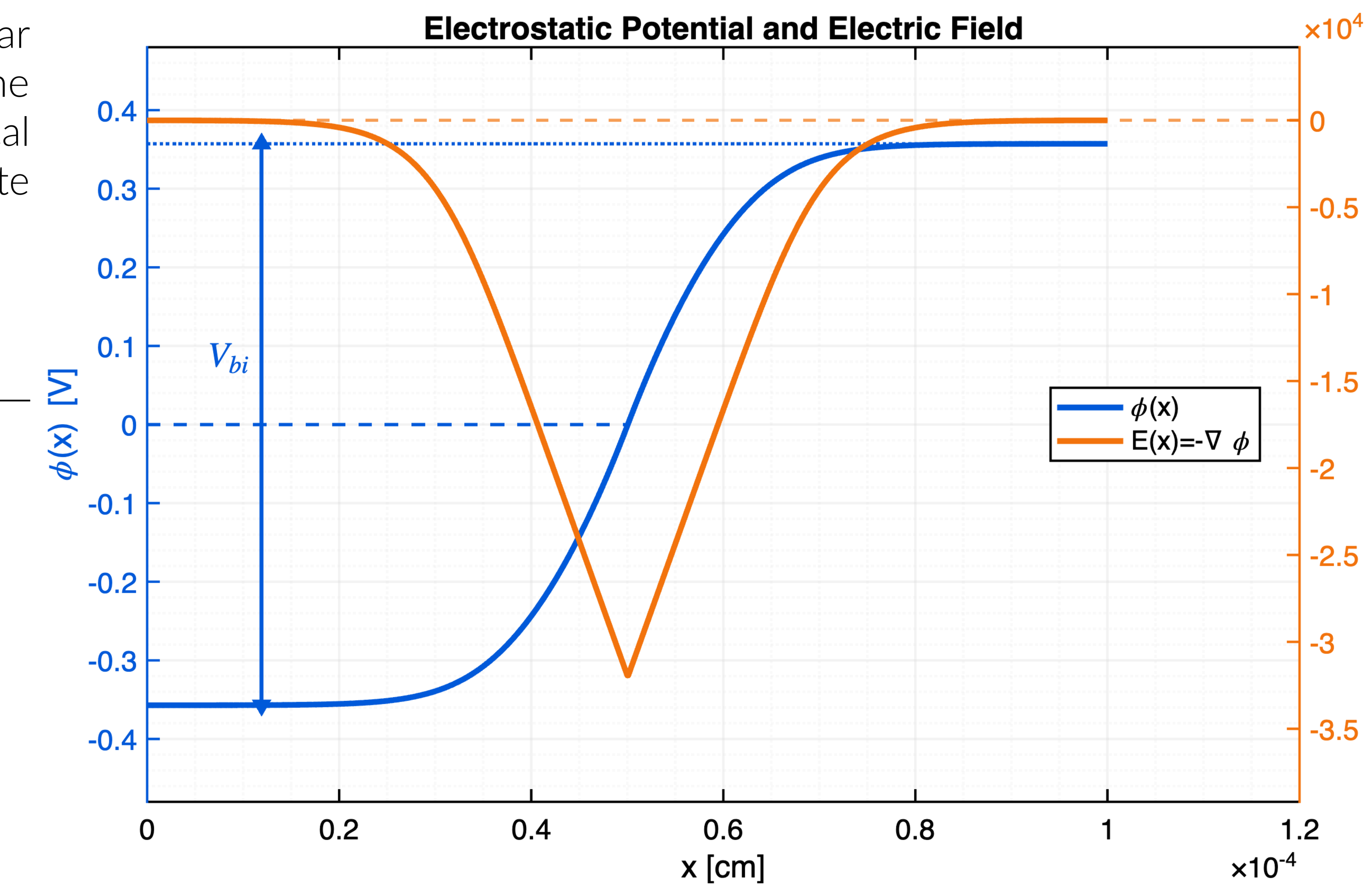
$$A\phi^{(k+1)} - d^{(k)}\phi^{(k+1)} = b(\phi^{(k)}) - d^{(k)}\phi^{(k)}, \qquad d_i^{(k)} := \frac{\lambda}{V_T}\cosh\left(\frac{\phi_i^{(k)}}{V_T}\right)$$

Matrix is ***strictly diagonally dominant*** *(equilibrium unchanged).*

Resulting Linear System:

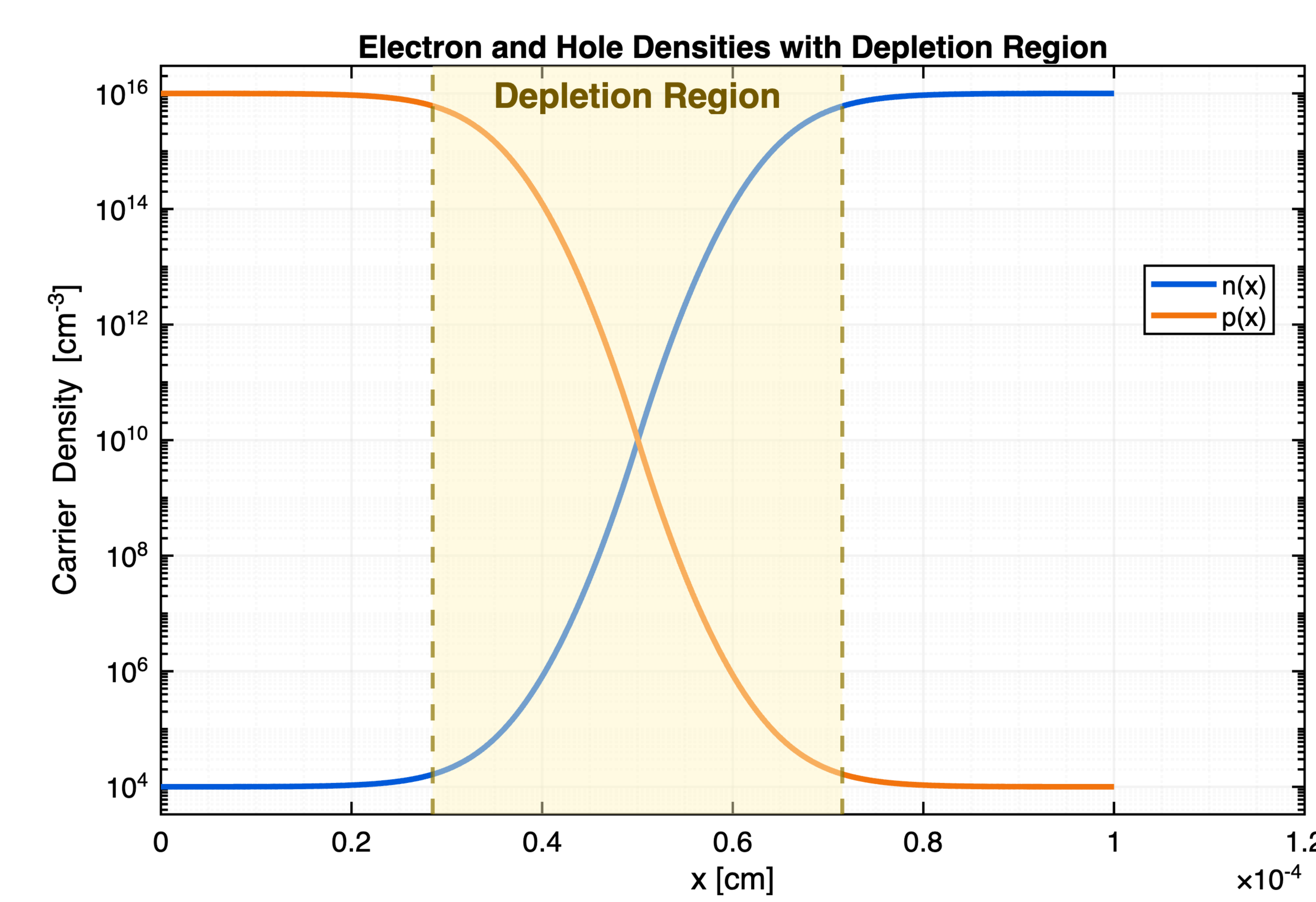
$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 - h^2 d_1^{(k)} & 1 & \cdots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 1 & -2 - h^2 d_{m-1}^{(k)} & 1 \\ 0 & \cdots & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_0^{(k+1)} \\ \phi_1^{(k+1)} \\ \vdots \\ \phi_{m-1}^{(k+1)} \\ \phi_m^{(k+1)} \end{bmatrix} = \begin{bmatrix} V_T \ln\left(\frac{n_i}{|C_0|}\right) \\ h^2 b[\phi^{(k)}]_1 - h^2 d_1^{(k)}\phi_1^{(k)} \\ \vdots \\ h^2 b[\phi^{(k)}]_{m-1} - h^2 d_{m-1}^{(k)}\phi_{m-1}^{(k)} \\ V_T \ln\left(\frac{C_m}{n_i}\right) \end{bmatrix}$$

Electrostatic Potential and Electric Field



The electric field is obtained from the potential through $\mathbf{E} = -\nabla\phi$. Flat regions of $\phi(x)$ correspond to $|\mathbf{E}| \approx 0$, while large gradients in $\phi(x)$ produce strong electric fields.

Carrier Densities and Depletion Region



The depletion region is centered near the junction, where electron and hole densities transition from p-type to n-type behavior.

Outlook

Extension to non-equilibrium drift–diffusion models:

$$-\nabla \cdot (\varepsilon \nabla \phi) = q(p - n + C)$$

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot J_n - R(n,p), \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \cdot J_p - R(n,p)$$